24 A straight wire XY 2.0 cm long carrying a current of 1.5 A is placed at the centre of a solenoid 1.2 m long with 20 000 turns of wire and carrying a current of 3.0 A, so that the wire XY is perpendicular to the magnetic field produced by the solenoid.

What is the magnetic force acting on the wire XY?

A 6.0×10^{-4} N **B** 7.2×10^{-4} N
C 1.9×10^{-3} N **D** 2.3×10^{-3} N